ASSHU – KASESE JOINT EXAMINATIONS BOARD (AKJEB)

MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

P252/2 CHEMISTRY — MARKING GUIDE

Aim

An investigation to determine the activation energy, Ea, of the reaction between potassium manganate (VII) and oxalic acid by varying temperature of the reaction mixture while recording time taken for the reaction to reach completion, so as to determine the suitable method for water and waste-water treatment.

Key area

- *A learner identifies aim with all key words*
- *Purpose*
- *Variables*

Basis code A_2

Weighted score = 05 (A=05)

- States the aim of the experiment with some key words A = 03
- No aim or incorrect aim A = 0

(b) Hypothesis

The activation energy of the reaction between potassium manganate (VII) and oxalic acid is more than 10,000 J mol^{-1} by varying temperature of the reaction mixture while recording the time taken for the reaction to reach completion.

Basis code: H

- States relevant hypothesis with all key words H = 06
- States a partially correct hypothesis with some key words H = 05
- Hypothesis is just one statement quoting a guessed value/range H = 04
- No or wrong hypothesis H = 0

(c) Variables

Independent variable: Temperature of the reaction mixture.

Dependent variable: Time taken for the purple solution to turn colourless.

Controlled variables: Volume of BA_1, BA_2 and sulphuric acid used.

Basis code: V

- States clearly all the three variables — V = 06
- States clearly only two variables — V = 05
- States clearly only one variable — V = 04
- No variable — V = 0

(d) Procedure of the experiment

(i) 25 cm^3 of BA_2 was run from a burette into a conical flask, followed by 10 cm^3–25 cm^3 of BA_2 and heated to 80°C.
(ii) 25 cm^3 of BA_1 was added to the hot solution and simultaneously the stopwatch was started.
(iii) The flask was shaken to mix and left to stand.
(iv) The time, t, taken for the solution to turn colourless was noted and recorded.
(v) Steps (i) to (iv) were repeated for temperatures of 70°C, 60°C, 50°C, 40°C and room temperature.
(vi) The results were recorded in the data table, including: value of temperature in Kelvin, (T = x + 273), log t, 1/T

Basis code: P_3

- States relevant and coherent procedure with relevant materials — P = 04
- States relevant incoherent procedure with relevant materials — P = 02
- Lists relevant materials — P = 01
- No relevant materials or irrelevant procedure — P = 0

(d) Risks and mitigations

- Cuts from broken glassware such as conical flasks and burettes which can cause injuries and pain. This can be mitigated by placing the glassware on a flat surface and wearing gloves and shoes to prevent direct contact with skin.
- Hot flames and solutions can cause scalds and burns when exposed to the skin. This can be mitigated by putting on a lab coat, gloves and closed shoes.

Basis code: R

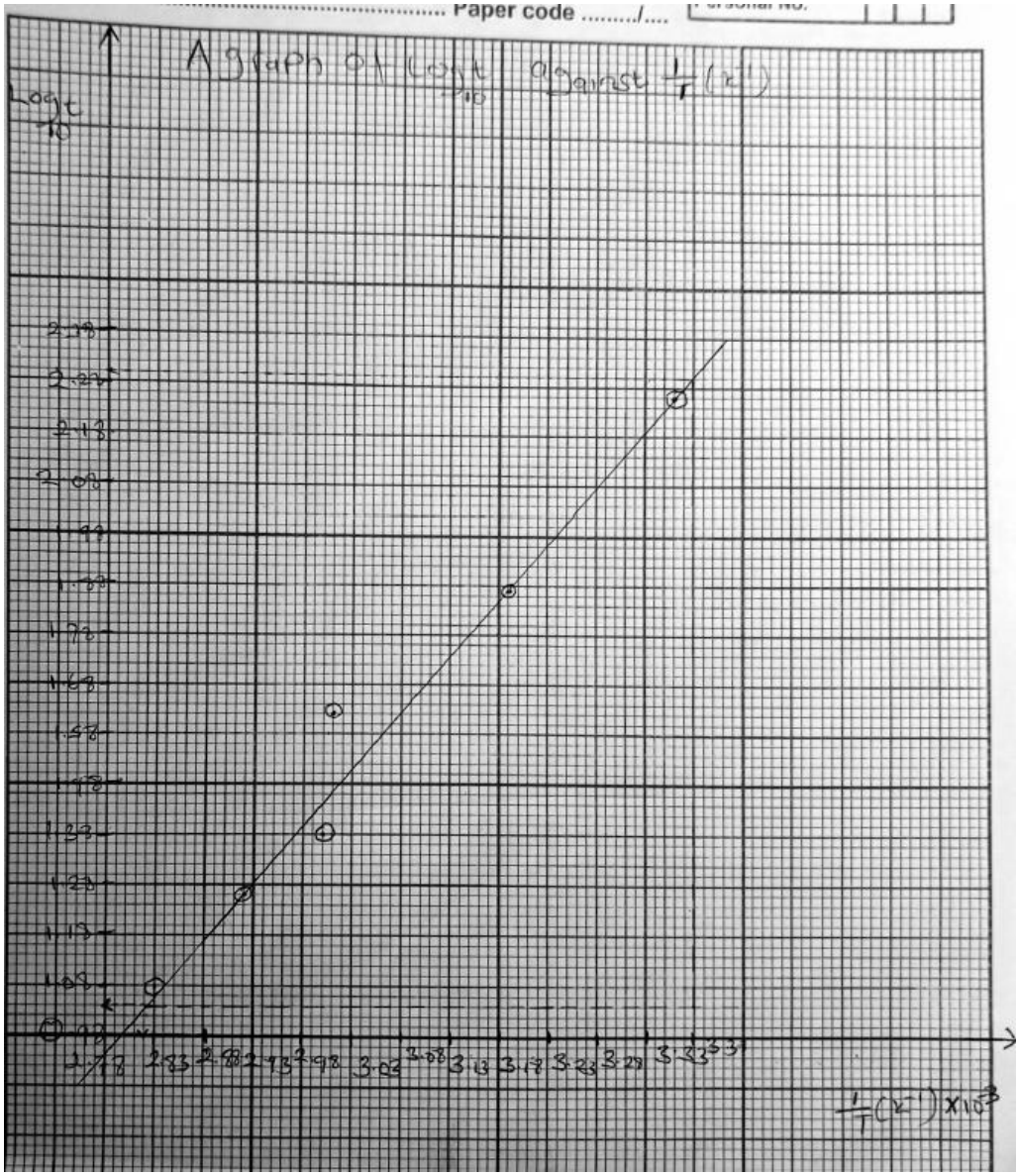
- Identifies one possible risk of the experiment and mitigation — R = 03
- No mitigation or any possible risk identified in the experiment — R = 0

(E) Data presentation and recording

Temperature / °C	**80.0**	**70.0**	**60.0**	**50.0**	**40.0**	**RT: 245**
Temperature, T/K	353.0	343.0	333.0	323.0	313.0	297.0
$1/T\ (K^{-1}) \times 10^3$	2.83	2.92	3.00	3.10	3.19	3.36
Time, t (s)	12.0	18.0	24.0	42.0	74.0	180.0
log t	1.08	1.26	1.38	1.62	1.87	2.26

Data presentation

- Appropriately presents not less than 6 sets of data — D = 05
- Data must be in a table and all variables presented with units.



Inappropriately presents not less than 6 sets of data — D = 03
Data in a suitable table but units missing.

Data recording

- Appropriately records the required decimal places/significant figures with correct trend — DR = 05
- Appropriately records the required decimal places/significant figures but outside the trend — DR = 03
- Correct trend e.g at 80^0 time should be from (7 – 12) seconds.
- No score for the values below 7 and above 12 seconds.

(F) Data Analysis and Interpretation

A graph of: $\log_{10} t$ against $(\frac{1}{T})$ (K^{-1})
From the graph:

$$k = \frac{change\ in\ \log_{10} t}{change\ in\ (\frac{1}{T})}$$

Using two points:

$$k = \frac{2.3-1.04}{(3.38 - 2.82) \times 10^{-3}} = 2250K$$

But slope: $k = \frac{Ea}{2303R}$, where R is the molar gas constant.
Therefore:

$Ea = (2250 \times 2.303 \times 8.314)\ J\ mol^{-1}$
$Ea = 43081.069\ J\ mol^{-1}$

Therefore, the activation energy for the reaction between potassium manganate (VII) and oxalic acid is $43081.069\ J\ mol^{-1}$

Conclusion and Recommendation

Activation energy, Ea, is $43081.069\ J\ mol^{-1}$ which is above 10,000 $J\ mol^{-1}$.
Therefore, the method is feasible and hence can be used for water and wastewater treatment.

Total Scores on Item 1

A = 05
H = 06
V = 06
P = 04
R = 03
D = 05
DR = 05
I = 11
C = 05

TOTAL = 50

POSSIBLE RESPONSES

Test	Observation	Deduction
(a) A spatula end full of W was put in a dry test tube, gently and then heated strongly using a Bunsen burner (heat source).	White solid. Colourless gas turns moist blue litmus paper red and lime water milky. Colourless gas which turns white anhydrous copper(II) sulphate to blue. White residue *Deny solid.*	Non-transition metal ions *(also accept ions)* CO_2, CO_3^{2-}, $C_2O_4^{2-}$ present. Water of crystallization present. CaO /Al_2O_3 hence Ca^{2+}/ Al^{3+} present. **08**
(b) A spatula end full of W was dissolved in 5 cm^3 water in a boiling tube, shaken and then filtered. Both filtrate and residue were kept.	White solid partially dissolves forming a white suspension. Colourless filtrate. White residue.	Ca^{2+}, Mg^{2+}, Ba^{2+}, Zn^{2+}, Al^{3+} are present. **04** *Accept words, e.g. aluminium ion* *Deny if the word ion is missing.*
(c) The filtrate was divided into five, (i) To the first portion, In a test tube, dilute sodium hydroxide solution was added dropwise until in excess.	White precipitate, soluble in excess forming a colourless solution. *Deny: if the word soluble is missing*	Zn^{2+}or Pb^{2+}or Al^{3+} present **02**
(ii) To the second part in a test tube, dilute ammonia solution was added dropwise until in excess.	White precipitate insoluble in excess. *Deny: if the word insoluble misses.*	Pb^{2+} or Al^{3+}. **02**
(iii) To the third portion, 3 drops of potassium iodide solution were added.	No observable change	Pb^{2+} absent. Al^{3+} present. **02** *Deny a score if Pb^{2+} absent misses*
(iv) Confirmatory test: Using the fourth portion in a test tube, 3 drops of dilute nitric acid were added followed by 3–4 drops of litmus solution and then dilute ammonia solution dropwise until in excess.	Blue lake solution.	Al^{3+} confirmed. **02** *Deny blue lake only*

(v) Fifth portion: Three drops of lead(II) nitrate solution were added and then heated.	White precipitate insoluble on heating. *Note: condition must be clearly stated.*	SO_4^{2-}. **02**
(vi) Confirmatory test: Using the sixth portion in a test tube, 4 drops of barium nitrate solution were added followed by dilute nitric acid.	White precipitate insoluble in acid.	SO_4^{2-}. **02**
(d) The residue in a test tube was washed with water, then dilute nitric acid was added to dissolve it. The resultant solution was divided into three portions in different test tubes.	White residue dissolves to form a colourless solution. Bubbles of a colourless gas which turns moist blue litmus paper red and lime water milky.	Ca^{2+}, Mg^{2+}, Zn^{2+}, Pb^{2+}, Al^{3+}. CO_2 gas given off. CO_3^{2-} confirmed. **04**
(i) To the first portion in a test tube, dilute sodium hydroxide solution was added dropwise until in excess.	White precipitate insoluble. Deny: If the technical term is misspelt, word insoluble misses.	Ba^{2+}, Ca^{2+}, Mg^{2+}. **02**
(ii) To the second portion in a test tube, dilute ammonia solution was added dropwise until in excess.	No observable change.	Ca^{2+}. **02**
(iii) Confirmatory test: To the third portion, ammonium oxalate solution followed by dilute hydrochloric acid was added.	White precipitate soluble in dilute nitric acid.	Ca^{2+} confirmed present. **02**
Note: A candidate may dissolve the solid in nitric acid followed by sodium hydroxide solution as procedures (b), (c) were repeated. If nitric acid. The other procedure must follow when nitric acid was used or not.		
(b) A spatula end full of W was dissolved in dilute nitric acid followed by sodium hydroxide dropwise until in excess. The solution was kept aside and cooled to room temperature.	Bubbles of colourless gas that turns blue litmus paper red and lime water milky. It dissolves to form a colourless solution. White precipitate insoluble in excess sodium hydroxide. Colourless filtrate. White residue.	CO_2 given off; CO_3^{2-} confirmed; Ba^{2+}, Ca^{2+}, Al^{3+}, Zn^{2+}. Ba^{2+}, Ca^{2+}, Mg^{2+}. Zn^{2+}, Al^{3+}, Pb^{2+}. Ba^{2+}, Ca^{2+}, Mg^{2+}.
(c) Nitric acid was added to filtrate and divided into six portions	White precipitate dissolves to form a colourless solution.	Zn^{2+}, Al^{3+}, Pb^{2+}.

(i) To the first portion, Sodium hydroxide was added dropwise until in excess.	White precipitate soluble in excess sodium hydroxide.	Zn^{2+}, Al^{3+}, Pb^{2+}.
(ii) To the second portion, Ammonia solution was added until in excess.	Refer to the first table	Refer to the first table
(iii) To the third portion, Potassium iodide (2 drops) was added.	Refer to the first table	
Confirmatory test: Use from the first procedure.		

Cations present are: Al^{3+}, Ca^{2+}

Anions present are: CO_3^{2-} and SO_4^{2-}

Application of calcium carbonate and aluminium sulphate in a cement factory

Calcium carbonate

Calcium carbonate provides the raw material calcium oxide for a clinker.

In the factory kiln at about 1450°C, it undergoes calcination:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

The calcium oxide then reacts with Silicon(IV) dioxide to form the four main clinker minerals:

C_3S, C_2S, C_3A, C_4AF

These give cement its strength.

Aluminium sulphate

Aluminium sulphate in a cement factory acts as a set accelerator/early strength agent added in small doses to cement or concrete.

Aluminium ions speed up hydration of C_3A and ettringite formation.

TOTAL = 40